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A Darboux-type Theory of Integrability for Discrete Dynamical Systems

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There is a method for searching first integrals for polynomial ordinary differential equations (usually called Darboux method) based on the knowledge of several of their invariant algebraic hypersurfaces. We extend this method to discrete dynamical systems, providing a way of searching invariants for them and we give several examples of application.

Keywords: Discrete dynamical system; Invariants; Darboux integrability; Differential equations

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INTRODUCTION

Consider a discrete dynamical system (DDS in short)

$$x_{n+1} = F(x_n), \quad (1)$$

where $F : \mathcal{U} \subset \mathbb{R}^m \rightarrow \mathbb{R}^m$, being $\mathcal{U}$ an open subset. We will say that H is an *invariant*[‡] for Eq. (1), if H is a real valued function defined in $\mathcal{V}$, an open and dense subset of $\mathcal{U}$, satisfying

$$H(F(x)) = H(x), \quad \text{for all } x \in \mathcal{V}.$$

As far as we know there are two different theories called Darboux theory of integrability. The first one concerns with second-order hyperbolic partial differential equations, while the second one deals with polynomial ordinary differential equations (PODE). This second theory gives a method for searching first integrals for PODE, see for instance Ref. [5] for a survey or Ref. [2] for an application for $m = 3$. In this paper, we adapt to DDS this second theory. Although, as it becomes apparent from our results, this adaptation can be done for any m , we will mainly fix our attention to $m = 2$.

Before describing our results we would like to comment that although for most DDS there is no way of finding an explicit invariant, the cases for which an invariant can be found have been extensively studied and they are relevant in the applications.

This paper is organized as follows: in the next section, we give some preliminary definitions as well as a remainder of the main ideas involved in the Darboux theory of integrability for PODE. The third section contains the main results of the paper: a new method for searching invariants for DDS (see Theorem 5 and Method 7) and several examples of application. Most of these examples are obtained from the literature. Among them we can quote: linear systems, the Lyness map, maps related to the arithmetic–geometric map and some rational maps. A nice property of our method is that it allows to present all the examples from a unified viewpoint. We also include two representative examples for which such an invariant exists but it cannot be found with the method that we present, just to give the idea that as in the case of PODE this method is not universal. We conclude third

[‡]Sometimes H is also called a first integral of the DDS.

section with an application of the knowledge of the invariants to the determination, in terms of the initial condition, of the limit of some DDS.

The last section will be devoted to relate the Darboux method for PODE with the one presented in this paper.

DEFINITIONS AND PRELIMINARY RESULTS

Let us recall the basic ideas of Darboux Theory of integrability for planar polynomial differential equations: consider the PODE $(\dot{x}, \dot{y}) = X(x, y)$, being X a polynomial vector field of degree n . It is said that an algebraic curve $R(x, y) = 0$ is an *invariant algebraic curve* for it if $\langle \nabla R(x, y), X(x, y) \rangle = K(x, y)R(x, y)$, for some polynomial K , called *the cofactor*. Notice that $R(x, y)$ has degree at most $n - 1$. Here $\langle \cdot, \cdot \rangle$ denotes the usual scalar product.

Exponential factors, complex curves, and other extensions are considered in the general theory, see Refs. [5,6], but we restrict our attention to the simplest situation. The Darboux Method for PODE is summarized in the following result:

THEOREM 1 *Let*

$$\begin{cases} \dot{x} = P(x, y), \\ \dot{y} = Q(x, y), \end{cases} \quad (2)$$

be a polynomial differential system of degree n . Then the following statements hold:

- i) *If $R_i(x, y) = 0$ for $i = 1, \dots, [(n(n+1))/2] + 1$ are different invariant irreducible algebraic curves for Eq. (2), with cofactors $K_i(x, y)$, respectively, then the differential equation has a Darboux first integral of the form*

$$H(x, y) = \prod_{i=1}^{\frac{n(n+1)}{2}+1} |R_i(x, y)|^{\alpha_i},$$

for some $\alpha_i \in \mathbb{R}$.

- ii) *If Eq. (2) has s invariant algebraic curves $R_i(x, y) = 0$, $i = 1, \dots, s$, such that there exist $\alpha_i \in \mathbb{R}$, $i = 1, \dots, s$ satisfying $\sum_{i=1}^s \alpha_i K_i(x, y) = 0$,*

being $K_i(x, y)$ the cofactors of each $R_i(x, y)$, then the differential equation has the Darboux first integral

$$H(x, y) = \prod_{i=1}^s |R_i(x, y)|^{\alpha_i}.$$

As we will see, in order to study DDS we shall modify statement (ii) of the above result (see Theorem 5 and Method 7). Notice that in the statement (ii), the fact that both $R_i(x, y)$ and $K_i(x, y)$ are polynomials, is not essential. On the other hand, the proof of part (i) is based on two facts: The cofactors are polynomials of degree at most $n - 1$, and if the system has enough of them (that is $\lfloor (n(n+1))/2 \rfloor + 1$) they must be linearly independent.

Let us introduce some definitions to state our method for searching invariants for DDS.

DEFINITION 2 Let V be an open subset of $\mathbb{R}^2$. Consider a map $R : \mathcal{V} \rightarrow \mathbb{R}$, not necessarily algebraic, and such that $\{(x, y) \in \mathbb{R}^2 : R(x, y) = 0\} \neq \emptyset$. We will say that the function $R(x, y)$ induces an invariant curve for the DDS, $(x_{n+1}, y_{n+1}) = F(x_n, y_n)$ if

$$R(F(x, y)) = K(x, y)R(x, y), \quad \text{for all } (x, y) \in \mathcal{B}, \quad (3)$$

where $\mathcal{B}$ is an open and dense subset of $\mathcal{V}$ and $\{R(x, y) = 0\} \cap \mathcal{B} = \emptyset$. The function K is called a cofactor of R with respect to F .

Remark 3 In the above definition we say that $R(x, y)$ induces an invariant curve because if a point $(x, y) \in \{R(x, y) = 0\}$ then $F(x, y) \in \{R(x, y) = 0\}$, i.e. the curve is an invariant curve for the DDS generated by F .

DEFINITION 4 When a function $R(x, y)$ satisfies Eq. (3) but $\{(x, y) \in \mathbb{R}^2 : R(x, y) = 0\} = \emptyset$ we will say that R is an exponential factor of F with cofactor K . For instance any function of the form $R(x, y) = \exp\{\varphi(x, y)\}$ is an exponential factor with $K(x, y) = \exp\{\varphi(F(x, y)) - \varphi(x, y)\}$. See Ref. [5] for a motivated introduction of the role of the exponential factors for PODE.

MAIN RESULTS AND EXAMPLES

This section is devoted to present our method for searching invariants for DDS as well as to apply it to several two dimensional examples and to one four dimensional example. We also include two classical DDS for which the method that we develop does not work. We conclude with an application of the knowledge of the invariants for some difference equations to determine their limiting behavior in terms of the initial condition.

The New Method

A first step to obtain the new method that we present in this paper is the following result:

THEOREM 5 *Let*

$$(x_{n+1}, y_{n+1}) = F(x_n, y_n), \quad (4)$$

be a discrete dynamic system defined in $\mathcal{U} \subset \mathbb{R}^2$. Let $R_i(x, y)$, $i = 1, 2, \dots, s$ be either functions inducing invariant curves or exponential factors of Eq. (4) with cofactors $K_i(x, y)$, $i = 1, \dots, s$, respectively. Then if there exists $\alpha_1, \alpha_2, \dots, \alpha_s \in \mathbb{R}$ such that

$$\prod_{i=1}^s |K_i(x, y)|^{\alpha_i} = 1 \quad (5)$$

in an open and dense subset of $\mathcal{U}$ then

$$H(x, y) := \prod_{i=1}^s |R_i(x, y)|^{\alpha_i},$$

is an invariant of Eq. (4).

Proof Since for each $i = 1, \dots, s$, we have that $R_i(F(x, y)) = K_i(x, y)R_i(x, y)$ in an open dense subset $\mathcal{B}_i$ of $\mathcal{U}$, it follows that

$$H(F(x, y)) = \prod_{i=1}^s |R_i(F(x, y))|^{\alpha_i} = \prod_{i=1}^s |K_i(x, y)|^{\alpha_i} \prod_{i=1}^s |R_i(x, y)|^{\alpha_i}.$$

Using Eq. (5) we obtain $H(F(x, y)) = H(x, y)$ for any $(x, y) \in \cap_{i=1}^s \mathcal{B}_i$, which is an open and dense subset of $\mathcal{U}$, as desired. $\square$

Remark 6 Note that if instead of obtaining a collection of α_i such that condition (5) holds, we get $\prod_{i=1}^s |K_i(x, y)|^{\alpha_i} = a \neq 1$ for some constant a , we have obtained a kind of non-autonomous invariant. This is because defining $H_n := H(x_n, y_n)$ we have that $H(F(x, y)) = aH(x, y)$, that is $H_{n+1} = aH_n$, or in other words $H_{n+1} = a^{n+1}H_0$.

A main difference between invariants curves for ODE and functions inducing invariant curves for DDS is the following: Assume that $R(x, y) = T(x, y)S(x, y) = 0$ is an invariant solution for an ODE. It is clear then that both $S(x, y) = 0$ and $T(x, y) = 0$ must be invariants for the same ODE. On the other hand, assume that $R(x, y) = T(x, y)S(x, y)$ induces an invariant curve for the DDS, $(x_{n+1}, y_{n+1}) = F(x_n, y_n)$, being both sets $\{S(x, y) = 0\}$ and $\{T(x, y) = 0\}$ nonempty. Then maybe $S(x, y)$ and $T(x, y)$ are as well functions which induce invariant curves for the DDS, but it also may occur that the DDS maps the invariant set $\{S(x, y) = 0\}$ into $\{T(x, y) = 0\}$, and vice versa. In other words, it may happen that both functions $S(x, y)$ and $T(x, y)$ induce invariants just for $F^2 = F \circ F$.

Finally, notice that the singularities of the map F that defines the DDS, as well as the preimages of all these singularities should also play an especial role in the dynamics generated by F . Also observe that if the singularities of F are contained in the set $\bigcup_{k=1}^l \{G_k(x, y) = 0\}$ then the preimages for F of these singularities are included in the sets $\{G_k \circ F^j(x, y) = 0\}$, varying $j \in \mathbb{N}$ and $k = 1 \dots l$.

The above comments and the results of Theorem 5 lead us to propose the following method for searching invariants for DDS:

METHOD 7 (A METHOD FOR SEARCHING INVARIANTS FOR DDS) *Let*

$$(x_{n+1}, y_{n+1}) = F(x_n, y_n), \quad (6)$$

be a DDS defined in $\mathcal{U} = \mathbb{R}^2 \setminus \bigcup_{k=1}^l \{G_k(x, y) = 0\}$ an open, dense subset of $\mathbb{R}^2$. Let $R_i(x, y)$, $i = 1, 2, \dots, s$ be functions inducing invariant curves for F^{k_i} for some $k_i \geq 1$, or exponential factors of F . Then a natural candidate to be an invariant for Eq. (6) is

$$H(x, y) := \prod_{i=1}^s |R_i(x, y)|^{\alpha_i} \prod_{j=0}^m \left(\prod_{k=1}^l |G_k \circ F^j(x, y)|^{\beta_{j,k}} \right), \quad (7)$$

where m is any given fixed natural number and the constants α_i and $\beta_{j,k}$ are real numbers to be determined.

As we will see in the next subsections, Method 7 together with Theorem 5 will provide an useful way of finding invariants for several DDS.

The Lyness Map

An invariant for the well-known Lyness map

$$F(x, y) = \left(y, \frac{a+y}{x} \right),$$

is already known [8]. Here we find it by using our method.

In the next table, we show all the linear functions which induce invariant lines for F , F^2 and F^3 as well as the line of singularities of F , $\{x = 0\}$, with some of its preimages.

$x + y + 1 = 0$;	Invariant under F^2
$x + 1 = 0$, $y + 1 = 0$, and $x + y + a = 0$;	Invariant under F^3
$x = 0$, $y = 0$ and $y + a = 0$;	Some preimages of $\{x = 0\}$.

Therefore, Method 7 provides a candidate to be an invariant of the form

$$H(x, y) = |x + y + 1|^{\alpha_1} |x + 1|^{\alpha_2} |y + 1|^{\alpha_3} |x + y + a|^{\alpha_4} |x|^{\beta_1} |y|^{\beta_2} |y + a|^{\beta_3}.$$

By imposing that $H(F(x, y)) = H(x, y)$ we get the solutions

$$\beta_3 = 0, \quad \alpha_1 = 0, \quad \alpha_2 = \alpha_3 = \alpha_4 = -\beta_1 = -\beta_2.$$

Taking $\alpha_2 = 1$, we obtain that an invariant for the Lyness map is

$$H(x, y) = \frac{(x+1)(y+1)(x+y+a)}{xy}.$$

In fact, there is a generalization of the Lyness map and of its invariant to higher dimensions, see Ref. [8]. This invariant also can be found by using our method.

Maps Related with the Arithmetic–Geometric Map

Gauss in 1799 and 1818, see Ref. [7], studied the map $F(x, y) = ((x+y)/2, \sqrt{xy})$ that combines the arithmetic and geometric means.

In “Two Examples of DDS for Which Our Method Does Not Work” section, we recall some of the Gauss’ results. In this section, we study two related maps which are also considered in Ref. [3]. Here, we give a different deduction of their invariants by using Theorem 5.

Firstly, for x and y positive, consider

$$F(x, y) = \left(\sqrt[p]{\frac{x(x^p - y^p)}{p(x - y)}}, \sqrt[p]{\frac{y(x^p - y^p)}{p(x - y)}} \right), \quad (8)$$

where $p \in \mathbb{N}, p \geq 2$. From direct inspection it is clear that $R_1(x, y) = y^p - x^p$ and $R_2(x, y) = \ln(y/x)$ induce invariants curves for the DDS generated by F . In fact,

$$R_1(F(x, y)) = \frac{1}{p} R_1(x, y) \quad \text{and} \quad R_2(F(x, y)) = \frac{1}{p} R_2(x, y).$$

Therefore, since their cofactors coincide, by Theorem 5, we obtain that the function

$$H(x, y) = \frac{R_1(x, y)}{R_2(x, y)} = \frac{y^p - x^p}{\ln y - \ln x},$$

is an invariant of Eq. (8).

Note that for $p = 2$, Eq. (8) gives the map

$$F(x, y) = \left(\sqrt{x \frac{x+y}{2}}, \sqrt{y \frac{x+y}{2}} \right), \quad (9)$$

which is similar to the map studied by Gauss.

As a second example consider, also for x and y positive and for instance $0 < x \leq y$, the map

$$F(x, y) = (F_1(x, y), F_2(x, y)) = \left(\frac{x+y}{2}, \sqrt{y \frac{x+y}{2}} \right).$$

This map was introduced by Gauss in 1800 in a letter to Pfaff and was also studied later by Borchardt.

Taking $R_1(x, y) = y^2 - x^2$ (as above when $p = 2$) we get a first function that induces an invariant curve, because

$$R_1(F(x, y)) = \frac{1}{4} R_1(x, y).$$

A second function is more difficult to be found. Note that

$$\frac{F_1(x, y)}{F_2(x, y)} = \sqrt{\frac{x+y}{2y}} = \sqrt{\frac{1}{2} \frac{x}{y} + \frac{1}{2}},$$

and that the right hand side of the above formula is reminiscent of the right hand side of the well-known equality

$$\cos A = \sqrt{\frac{1}{2} \cos 2A + \frac{1}{2}}.$$

Hence it becomes natural to consider $\cos 2A = x/y$. Therefore, we obtain

$$\cos A = \frac{F_1(x, y)}{F_2(x, y)},$$

or equivalently,

$$\arccos\left(\frac{F_1(x, y)}{F_2(x, y)}\right) = \frac{1}{2} \arccos\left(\frac{x}{y}\right).$$

The above equality leads also to

$$R_2(F(x, y)) = \frac{1}{2} R_2(x, y),$$

being $R_2(x, y) = \arccos(x/y)$ a second function which induces a new invariant curve for Eq. (9). By using Theorem 5, we get that

$$H(x, y) = \frac{\arccos^2(x/y)}{y^2 - x^2},$$

in an invariant for Eq. (9).

Finally observe that in the study of both maps (8) and (9) we get constant co-factors. By using Remark 6, for instance in this second case, we get the following equalities associated to R_1 and R_2 :

$$y_n^2 - x_n^2 = \left(\frac{1}{4}\right)^n (y_0^2 - x_0^2),$$

and

$$\arccos^2\left(\frac{x_n}{y_n}\right) = \left(\frac{1}{2}\right)^n \arccos^2\left(\frac{x_0}{y_0}\right).$$

Notice that from the above two equalities, it is easy to get (x_n, y_n) in terms of n , x_0 and y_0 .

Linear Maps

In this section, we use our method to find invariants for DDS generated by planar linear maps with real eigenvalues. By performing an affine change of variables we can assume that $F(x, y) = A \cdot (x, y)^t$, where A is a 2×2 matrix of one of the following forms:

$$A_1 = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad \text{with } a, b \in \mathbb{R}.$$

and

$$A_2 = \begin{pmatrix} a & 0 \\ 1 & a \end{pmatrix}, \quad \text{with } a \in \mathbb{R}.$$

Let us construct an invariant in each one of the cases:

- (1) Case $A = A_1$; $F(x, y) = (ax, by)$. Taking $R_1(x, y) = x$ and $R_2(x, y) = y$ we get $R_1(F(x, y)) = aR_1(x, y)$, $R_2(F(x, y)) = bR_2(x, y)$. In order to apply Theorem 5 we search for α_1, α_2 such that $|a|^{\alpha_1}|b|^{\alpha_2} = 1$. We just consider the cases $a \cdot b \neq 0$, $|a| \neq 1$, or $|b| \neq 1$, because otherwise the dynamical system is trivial. Taking for instance $\alpha_1 = \ln |b|$ and $\alpha_2 = -\ln |a|$ we obtain the invariant

$$H(x, y) = |x|^{\ln |b|} |y|^{-\ln |a|}.$$

For Instance, consider the Fibonacci recurrence $f_{n+1} = f_n + f_{n-1}$. It is equivalent to the DDS generated by $F(x, y) = (y, x + y)$. The above reasoning lead us to the invariant

$$H(x, y) = |y - \alpha^+ x|^{\ln |\alpha^+|} |y - \alpha^- x|^{\ln |\alpha^-|},$$

where $\alpha^\pm = (1 \pm \sqrt{5})/2$ are the roots of $x^2 + x + 1 = 0$.

- (2) Case $A = A_2$; $F(x, y) = (ax, x + ay)$. We just consider $a > 0$. As in the previous case $R_1(x, y) = x$ induces an invariant curve because $R_1(F(x, y)) = aR_1(x, y)$.

On the other hand, it is not difficult to verify that $R_2(x, y) = \exp(y/x)$ is an exponential factor for F . Observe that

$$R_2(F(x, y)) = \exp\left(\frac{x + ay}{ax}\right) = \exp\left(\frac{1}{a}\right) \exp\left(\frac{y}{x}\right).$$

Therefore, by using Theorem 5 again, we get that

$$H(x, y) = |x| \exp\left((-a \ln a) \frac{y}{x}\right),$$

is an invariant of the linear map in the case $A = A_2$.

A Type of Rational Maps

In this section, we obtain an invariant for the rational map studied by Sahadevan in Ref. [11]:

$$F(x, y) = \left(y, \frac{P(x, y)}{Q(x, y)}\right),$$

where

$$P(x, y) = \begin{vmatrix} I_0(y) & I_1(y) \\ I_3(y) & I_4(y) \end{vmatrix} - \begin{vmatrix} I_2(y) & I_0(y) \\ I_5(y) & I_3(y) \end{vmatrix},$$

and

$$Q(x, y) = \begin{vmatrix} I_2(y) & I_0(y) \\ I_5(y) & I_3(y) \end{vmatrix} - \begin{vmatrix} I_1(y) & I_2(y) \\ I_4(y) & I_5(y) \end{vmatrix},$$

being

$$\begin{aligned} I_0(y) &= A_1 + A_2y + A_3y^2, & I_3(y) &= B_1 + B_2y + B_3y^2, \\ I_1(y) &= A_2 + A_5y + A_6y^2, & I_4(y) &= B_2 + B_5y + B_6y^2, \\ I_2(y) &= A_3 + A_6y + A_9y^2, & I_5(y) &= B_3 + B_6y + B_9y^2. \end{aligned}$$

By using a computer algebra package, we get that $R_1(x, y) = I_0(x) + I_1(x)y + I_2(x)y^2$, and $R_2(x, y) = I_3(x) + I_4(x)y + I_5(x)y^2$, induce

invariant curves for the DDS. More concretely

$$R_1(F(x, y)) = \frac{R_8(y)}{Q^2(x, y)} R_1(x, y),$$

and

$$R_2(F(x, y)) = \frac{R_8(y)}{Q^2(x, y)} R_2(x, y),$$

being $R_8(y)$ a polynomial of degree 8 in the variable y with coefficients in the variables A_i and B_i . Since both functions have the same cofactor, Theorem 5 implies that

$$H(x, y) = \frac{R_1(x, y)}{R_2(x, y)},$$

is an invariant of the DDS generated by the rational map.

An Example Constructed from Involutions

Consider the map

$$F(x, y) = \left(\frac{ay + b}{y - a}, \frac{ax + b}{x - a} \right), \quad (10)$$

where a and b are real numbers. By searching quadratic functions which induce invariant curves with a computer algebra package we get that $R(x, y) = b + a(x - y) + xy = 0$ is one of them. In fact,

$$R(F(x, y)) = \frac{b + a^2}{(x - a)(y - a)} R(x, y).$$

Furthermore, the curves $x - a = 0$ and $y - a = 0$ are singularities of the map F . Hence, by using Method 7, a good family of candidates to be an invariant for the DDS generated by F is

$$H_{\alpha, \beta, \gamma}(x, y) = |b + a(x - y) + xy|^\alpha |x - a|^\beta |y - a|^\gamma.$$

By imposing that $H(F(x, y)) = H(x, y)$, we get several solutions. In particular

$$H_{1,-1,0} = \frac{b + a(x - y) + xy}{x - a}, \quad \text{and} \quad H_{0,1,-1} = \frac{x - a}{y - a}$$

are independent invariants of the DDS generated by F .

Remark 8

- (a) Observe that the fact that the map Eq. (10) has two independent invariants determines completely the behavior of the dynamical system. This is not a possible situation in the case of planar ODE. If a planar system has two first integrals, then they are functionally dependent.
- (b) The map $\varphi(x) = (ax + b)/(x - a)$ is an involution ($\varphi(\varphi(x)) \equiv x$), and the structure of Eq. (10) is $F(x, y) = (\varphi(y), \varphi(x))$ which is also an involution. For these type of maps, it is easy to check that $H_1(x, y) = y + \varphi(x)$ and $H_2(x, y) = x + \varphi(y)$ are always invariants. In fact, H_1 is the invariant $H_{1,-1,0}$ obtained by using our method.

An Example in $\mathbb{R}^4$: The Cross-ratio as an Invariant of a DDS

Consider the homography $\varphi(x) := (ax + b)/(cx + d)$, and from it define the map $F : \mathcal{U} \subset \mathbb{R}^4 \rightarrow \mathbb{R}^4$, $F(x) = (\varphi(x_1), \varphi(x_2), \varphi(x_3), \varphi(x_4))$. If we take $R_{i,j}(x) := x_i - x_j$ one can verify that

$$R_{i,j}(F(x)) = \frac{\det A}{\varphi(x_i)\varphi(x_j)} R_{i,j}(x),$$

for all $i \neq j \in \{1, 2, 3, 4\}$, where

$$A := \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Hence the functions $R_{i,j}(x)$ induce the invariant hypersurfaces $R_{i,j}(x) = 0$ for the DDS generated by F , and their cofactors are $(\det A)/(\varphi(x_i)\varphi(x_j))$. If $\det A = 0$ then each $R_{i,j}(x)$ is an invariant for F . If $\det A \neq 0$ then our method given in Theorem 5 (extended to $\mathbb{R}^4$ in the natural way) gives that

for each choice of different $i, j, k, l \in \{1, 2, 3, 4\}$, the function

$$H(x) = \frac{R_{i,j}(x)R_{k,l}(x)}{R_{i,k}(x)R_{j,l}(x)}$$

is an invariant for the DDS generated by F . Note that the equality $H(F(x)) = H(x)$ can be interpreted as the conservation of the cross-ratio for homographies.

Two Examples of DDS for Which Our Method Does Not Work

The aim of this subsection is to clarify that although the method introduced in this paper works for several DDS, it is not an universal method for searching invariants.

Example A (The Arithmetic–Geometric Map, [3,7]) Consider x and y positive and

$$F(x, y) = \left(\frac{x+y}{2}, \sqrt{xy} \right). \quad (11)$$

As we already have commented, the above map was studied by Gauss. He proved that

$$H(x, y) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{x^2 \cos^2 \theta + y^2 \sin^2 \theta}}$$

is an invariant for the dynamic system defined by Eq. (11). Notice that the above expression is an elliptic integral. For a different proof of that of Gauss of the fact that $H(x, y)$ is an invariant of F , and a study of similar related maps, see again Ref. [7].

Example B (The Cohen Map, [10]) Consider the following map:

$$F(x, y) = \left(\sqrt{1+x^2} - y, x \right). \quad (12)$$

Although the numerical studies show exact integrability an explicit expression of an invariant for F is not known. In particular, in Ref. [10], the authors prove that it does not exist any algebraic invariant for Eq. (12), that is, an invariant map $H(x, y)$ such that $G(x, y, H(x, y)) = 0$ for some three variable polynomial G .

An Application of the Knowledge of an Invariant

Assume that for the DDS generated by F it is proved that $\lim_{n \rightarrow \infty} F^n(x_0, y_0)$ exists. The knowledge of an invariant for the DDS allows to get this limit in terms of (x_0, y_0) . We collect some examples of application of the above idea in next result.

PROPOSITION 9 *Consider the four DDS given by*

$$F_1(x, y) = (y, \mu x + (1 - \mu)y), \quad 0 < \mu < 1, \quad (x_0, y_0) \in \mathbb{R}^2,$$

$$F_2(x, y) = \left(\sqrt[p]{\frac{x(x^p - y^p)}{p(x - y)}}, \sqrt[p]{\frac{y(x^p - y^p)}{p(x - y)}} \right),$$

$$(x_0, y_0) \in (0, \infty) \times (0, \infty), \quad p \in \mathbb{N}, \quad p \geq 2,$$

$$F_3(x, y) = \left(\frac{x + y}{2}, \sqrt{y \frac{x + y}{2}} \right), \quad 0 < x_0 \leq y_0, \quad y_0 \in (0, \infty),$$

$$F_4(x, y) = \left(\frac{x + y}{2}, \sqrt{xy} \right), \quad (x_0, y_0) \in (0, \infty) \times (0, \infty),$$

then $\lim_{n \rightarrow \infty} F_j^n(x_0, y_0) = (\ell_j(x_0, y_0), \ell_j(x_0, y_0))$, $j = 1, 2, 3, 4$, where

$$\ell_1(x_0, y_0) = \frac{\mu x_0 + y_0}{\mu + 1}, \quad \ell_2(x_0, y_0) = \sqrt[p]{\frac{y_0^p - x_0^p}{p(\ln y_0 - \ln x_0)}},$$

$$\ell_3(x_0, y_0) = \sqrt{\frac{y_0^2 - x_0^2}{\arccos^2(x_0/y_0)}},$$

$$\ell_4(x_0, y_0) = \frac{\pi}{2} \left(\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{x_0^2 \cos^2 \theta + y_0^2 \sin^2 \theta}} \right)^{-1}.$$

Proof The DDS generated by $F_j(x, y)$, $j = 2, 3, 4$ are studied in “Maps Related with the Arithmetic–Geometric Map”, “An Example Constructed from Involutions” and “Two Examples of DDS for Which Our Method Does Not Work” sections, respectively, and an invariant $H_j(x, y)$ for each one of them is obtained. The DDS generated by $F_1(x, y)$ is linear. By using the results of “Linear Maps” section, we get the invariant $H_1(x, y) = \mu x + y$. In all cases it is not difficult to prove that $\lim_{n \rightarrow \infty} F_j^n(x_0, y_0)$ exists

and that it is of the form $(\ell_j(x_0, y_0), \ell_j(x_0, y_0))$. We will give the details for the case $j = 2$. The other cases follow with a similar study. Firstly observe that all the points of the form (x_0, x_0) are fix points for F_2 . Set $(x_n, y_n) = F_2(x_{n-1}, y_{n-1})$. Assume for instance that the initial condition satisfies $x_0 < y_0$. Therefore, it is easy to check that $x_0 < x_1 < y_1 < y_0$. Hence the sequence $\{x_n\}$ (resp. $\{y_n\}$) is monotone increasing (resp. decreasing) and bounded above (resp. below). So $\lim_{n \rightarrow \infty} (x_n, y_n)$ exists. By using the expression of F_2 it follows that the limit has to be of the form $(\ell_2(x_0, y_0), \ell_2(x_0, y_0))$, as we wanted to prove. In other words, we have to solve the system

$$H_2(x, y) = \frac{y^p - x^p}{\ln y - \ln x} = \frac{y_0^p - x_0^p}{\ln y_0 - \ln x_0}, \quad x = y.$$

By applying the L'Hôpital's rule we get the solution

$$x = y = \sqrt[p]{\frac{y_0^p - x_0^p}{p(\ln y_0 - \ln x_0)}} := \ell_2(x_0, y_0). \quad \square$$

Remark 10

- (1) In the case $j = 4$ of the above proposition the number $\ell_4(x_0, y_0)$ is called the arithmetic–geometric mean of x_0 and y_0 and has been already computed by Gauss[¶].
- (2) The case $j = 1$ of the above proposition corresponds also to the study of the second order linear difference equation $x_{n+1} = (1 - \mu)x_n + \mu x_{n-1}$. Note that the result proved implies that $\lim_{n \rightarrow \infty} x_n = (\mu x_0 + x_1)/(\mu + 1)$, and it is obtained without solving the difference equation.
- (3) Finally observe that all the results obtained in the above proposition correspond to the computation of some kind of generalized means of the numbers of x_0 and y_0 .

Remark 11 The following problem was proposed by Ladas several years ago, see also Ref. [1]: “Consider the second order difference equation

[¶]The fact that the iterates of F_4 are very easy to be computed and converge quadratically to their limit makes that this method is usually used to compute elliptic integrals.

$x_{n+1} = 1 + x_{n-1}/x_n$. It is known that every solution with initial condition x_1 and x_2 positive numbers tends to a period two solution located on the curve $D := \{(x, y) : y = x/(x - 1)\}$. Determine the two periodic solution in terms of the initial condition.”

The above problem can be reformulated in terms of $F^2 = F \circ F$, where $F(x, y) = (y, 1 + x/y)$: Consider the map $F^2(x, y) = ((x + y)/y, (x + y + y^2)/(x + y))$. As a consequence of the study of the difference equation we have that every solution with initial condition (x_1, y_1) , both positive numbers, tends to a fix point of F^2 , also located in $\mathcal{D}$. If we were able to find an invariant for the DDS generated by F^2 , arguing as in the proof of Proposition 9, the problem would be solved. The difficulty of this approach comes from the fact that the expression of invariants is not always given by elementary functions, as for instance, in the case of the arithmetic–geometric map studied by Gauss.

SOME RELATION BETWEEN DARBOUX METHOD OF INTEGRABILITY FOR PODE AND THE METHOD DEVELOPED IN THIS PAPER

To fix the relation we are searching, we proceed with a concrete example: Consider the differential system

$$\begin{cases} \dot{x} = -y + 4x^2y^2, \\ \dot{y} = x + 4xy^3. \end{cases} \quad (13)$$

(System H_4 of Ref. [4], p. 45). This system of differential equations has an isochronous center at the origin (this follows because in polar coordinates (r, θ) , it satisfies $\dot{\theta} \equiv 1$). We check that it has a first integral by using the Darboux method for PODE:

Set $X(x, y) = (-y + 4x^2y^2, x + 4xy^3)$. Consider $R_1(x, y) = x^2 + y^2$ and $R_2(x, y) = 1 + 4y^3$. Then one can check that

$$\langle \nabla R_1(x, y), X(x, y) \rangle = K_1(x, y)R_1(x, y), \quad \text{with } K_1(x, y) := 8xy^2,$$

and

$$\langle \nabla R_2(x, y), X(x, y) \rangle = K_2(x, y)R_2(x, y), \quad \text{with } K_2(x, y) := 12xy^2.$$

Therefore, $3K_1(x, y) - 2K_2(x, y) = 0$ and by Theorem 1. (ii) we obtain

$$H(x, y) = \frac{(x^2 + y^2)^3}{(1 + 4y^3)^2}, \quad (14)$$

is a first integral of system (13).

Consider now the flow associated to system (13). It is given by

$$\begin{aligned} \varphi(t, (x_0, y_0)) \\ = \left(\frac{x_0 \cos t - y_0 \sin t}{\sqrt[3]{1 + 4y_0^3 - 4(x_0 \sin t + y_0 \cos t)^3}}, \frac{x_0 \sin t + y_0 \cos t}{\sqrt[3]{1 + 4y_0^3 - 4(x_0 \sin t + y_0 \cos t)^3}} \right). \end{aligned}$$

Notice that in a neighborhood of the origin the above map is the identity for $t = 2\pi$ (that is, the flow is isochronous). Consider the flow at the time $t = \pi$. Then define

$$F(x, y) := \varphi(\pi, (x, y)) = \left(\frac{-x}{\sqrt[3]{1 + 8y^3}}, \frac{-y}{\sqrt[3]{1 + 8y^3}} \right). \quad (15)$$

The above map should have as invariant the first integral of Eq. (13), because Eq. (15) comes from the flow-map. Let us test that the method introduced in this paper also gives the invariant Eq. (14) by using as functions, which induce invariant curves the ones used for the PODE.

Take $R_1(x, y) = x^2 + y^2$ and $R_2(x, y) = 1 + 4y^3$. Then

$$R_1(F(x, y)) = \tilde{K}_1(x, y)R_1(x, y), \quad \text{with } \tilde{K}_1(x, y) := (1 + 8y^3)^{-2/3}$$

and

$$R_2(F(x, y)) = \tilde{K}_2(x, y)R_2(x, y), \quad \text{with } \tilde{K}_2(x, y) := (1 + 8y^3)^{-1}.$$

Notice that

$$\frac{\tilde{K}_1^3(x, y)}{\tilde{K}_2^2(x, y)} = 1,$$

and, therefore, by Theorem 5, Eq. (14) is an invariant for the DDS generated by Eq. (15).

Observe that in fact the map (15) has another independent invariant given by $H_2(x, y) = y/x$, and, as in Example defined by Eq. (10), both invariants fully determine the dynamical system generated by F .

Next result generalizes the relation between the integrability of PODE and DDS detected in the previous example:

THEOREM 12 *Let*

$$\begin{cases} \dot{x} = P(x, y), \\ \dot{y} = Q(x, y), \end{cases} \quad (16)$$

be a differential system, such that it has s invariant algebraic curves $R_i(x, y) = 0$, $i = 1, \dots, s$, with cofactors $K_i(x, y)$, such that there exist $\alpha_i \in \mathbb{R}$, $i = 1, \dots, s$ satisfying $\sum_{i=1}^s \alpha_i K_i(x, y) = 0$. Let $\varphi(t, (x_0, y_0))$ be the solution of Eq. (16) satisfying $\varphi(0, (x_0, y_0)) = (x_0, y_0)$. Consider $F_t(x, y) := \varphi(t, (x_0, y_0))$ for some $t \in \mathbb{R}$ for which the flow φ is well defined. Note that $\varphi: \mathcal{U} \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Then the following statements hold:

- i) The functions $R_i(x, y)$ also induce invariant curves for the DDS generated by a F_t .*
- ii) The cofactors of each $R_i(x, y)$ are $K_i^t(x, y) = \exp\left[\int_0^t K_i(\varphi(s, (x, y))) ds\right]$.*
- iii)*

$$\prod_{i=1}^s |K_i^t(x, y)|^{\alpha_i} = 1.$$

- iv) The PODE (resp. the DDS generated by F_t) has the function*

$$H(x, y) = \prod_{i=1}^s |R_i(x, y)|^{\alpha_i}$$

as a first integral (resp. invariant).

Proof We prove statements (i) and (ii) because the other ones follow straightforward from them. We shall prove them simultaneously. If $R(x, y) = 0$ is an invariant by Eq. (16) we have that $R(\varphi(t, (x, y))) = 0$ for all t in the interval of definition of the solution. Therefore,

$$R(\varphi(t, (x, y))) = K(t, (x, y))R(x, y), \quad (17)$$

for some $K(t, (x, y))$. Let us compute this $K(t, (x, y))$. Making derivatives with respect to t we get

$$\langle \nabla R(\varphi(t, (x, y))), f(\varphi(t, (x, y))) \rangle = \frac{\partial K(t, (x, y))}{\partial t} R(x, y).$$

Since $\{R(x, y) = 0\}$, is invariant for Eq. (16) with cofactor $K(x, y)$ we obtain

$$K(\varphi(t, (x, y)))R(\varphi(t, (x, y))) = \frac{\partial K(t, (x, y))}{\partial t} R(x, y).$$

By using Eq. (17) we get

$$\frac{\partial K(t, (x, y))}{\partial t} = K(\varphi(t, (x, y)))K(t, (x, y)),$$

or equivalently (ii) as we wanted to prove. Observe that in statement (ii) we have denoted $K(t, (x, y))$ by $K^1(x, y)$. $\square$

COROLLARY 13 *If a PODE can be integrated by the Darboux Method then any DDS defined by the flow of this PODE can also be integrated by the method proposed in this paper.*

Another example is given by the following isochronous system studied by Loud in [9],

$$\begin{cases} \dot{x} = -y + x^2, \\ \dot{y} = x(1 + y). \end{cases} \quad (18)$$

Its flow at time $\pi/2$ is

$$F(x, y) := \varphi\left(\frac{\pi}{2}, (x, y)\right) = \left(\frac{-y}{1 + y - x}, \frac{x}{1 + y - x}\right). \quad (19)$$

If we consider the functions $R_1(x, y) = x^2 + y^2$ and $R_2(x, y) = 1 + y$ we obtain by using either the Darboux Method or our method the first integral (or invariant),

$$H(x, y) = \frac{x^2 + y^2}{(1 + y)^2}.$$

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